

30 squarefree $\Rightarrow$ the seam μ_4 is not a hand ($4 \nmid 30$) but the carrier's Galois group

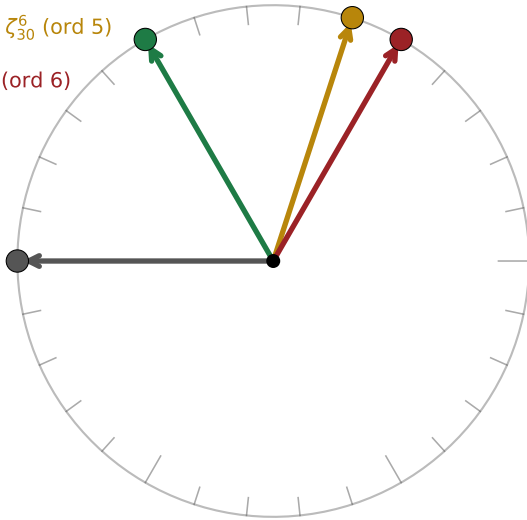
Cyclic dial $\langle \zeta_{30} \rangle$, order $h(E_8) = 30$

sheet $-1 = \zeta_{30}^{15}$ (ord 2)

family $\omega = \zeta_{30}^{10}$ (ord 3)

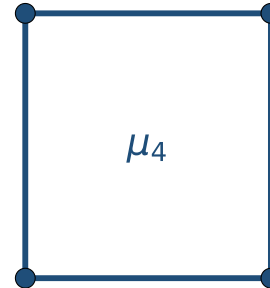
carrier $\zeta_5 = \zeta_{30}^6$ (ord 5)

CP $\zeta_6 = \zeta_{30}^5$ (ord 6)



Galois gears, order $\text{rank } E_8 = 8$

SEAM = $\text{Gal } \mathbb{Q}(\zeta_5) = (\mathbb{Z}/5)^\times$



$i: \zeta_5 \mapsto \zeta_5^2$ (Frobenius)

$\mathbb{Z}_2$



CP = $\text{Gal } \mathbb{Q}(\zeta_3) = (\mathbb{Z}/3)^\times$

$(\mathbb{Z}/30)^\times = \mu_4 \times \mathbb{Z}_2$, **order** $\varphi(30) = 8 = \text{rank } E_8$